Supervoxel-Based Intrinsic Scene Properties From Hyperspectral Images and LiDAR

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***Abstract*— The combination of spectral and 3-D elevation information provided by hyperspectral images (HSIs) and Light Detection and Ranging (LiDAR) has gained increased attention in the remote sensing field and enabled numerous applications. While various methods have been proposed to fuse these two data streams in pixel, feature, or decision level, a deeper view into the intrinsic relation of surface geometry, material reflectance, and environment illumination is still lacking. In this article, we present a novel supervoxel-based joint intrinsic decomposition framework for HSIs and LiDAR. First, we proposed a novel intrinsic scene model for HSIs and LiDAR point cloud, which tells how we can map LiDAR point cloud into HSI pixels with point-cloud-level normals, reflectance, and incident light direction. Then, we extract supervoxels from the LiDAR point cloud using a graph-based supervoxel method. Finally, we formulate the intrinsic decomposition problem within a supervoxel-based framework which can be optimized effectively and efficiently. The outputs of the proposed model are intrinsic scene properties like incident light direction and point-cloud-level hyperspectral reflectance, with which we can then generate intrinsic hyperspectral point cloud (IHSPC) where each point possesses not only 3-D coordinates and normals but also the reflectance over each wavelength. The performance of our approach is demonstrated with both synthetic and real data.**

***Index Terms*— Data fusion, hyperspectral images (HSIs), hyperspectral point cloud (HSPC), intrinsic hyperspectral image decomposition (IHID), light detection and ranging (LiDAR), supervoxel.**

I. Introduction

With the development of remote sensing technologies and imaging sensors, abundant land cover information can be acquired for earth observation. What is more, the simultaneous acquisition of multisensor data becomes more easily accessible due to the miniaturization of sensors which makes them easy to be integrated and mounted on unmanned aerial vehicles (UAVs) or airborne platforms. Surely collecting multimodal earth surface parameters enables numerous land cover mapping and environment applications, but how to combine or fuse the information provided by different data streams remains a big challenge. Of especial interest for various applications is the combination of spectral and 3-D elevation information provided by hyperspectral images (HSIs) and Light Detection and Ranging (LiDAR), respectively, as these two data souses are highly complementary to each other: hyperspectral imaging produces an image with rich spectral information, but the 3-D geometry structures of an object degenerate to a 2-D plane, whereas the LiDAR can make a dense 3-D laser scanning of ground surface and provide 3-D representations of the object, but its ability to discriminate different materials with intensity information is relatively weak. For this purpose, many HSIs and LiDAR fusion methods have been proposed, most of which conduct on a pixel, feature, or decision level. More specifically, the pixel-level methods transform point cloud into 2-D images like digital surface models (DSMs) and attach elevation information as new image band to HSIs [1], [2]; the feature-level methods are based on the pixel-level methods but further fuse geometric features from DSMs and spectral features from HSIs [3]–[6]; the decision-level run classifications over LiDAR and HSIs separately and then merge the classification label together [7], [8]. In general, the majority of current approaches transform and encapsulate 3-D point cloud into a 2-D image [9], which would facilitate calculation and analysis but also result in information loss. Moreover, the existing methods mainly make use of the heterogeneity of HSIs and LiDAR data, whereas ignoring the fact that spectral information is the result of the complex interplay of surface geometry, illumination, and material reflectance and thus has an inner-relation with LiDAR elevation information.

To preserve the irregular and full 3-D information contained in point clouds, instead of sampling point clouds into the image grid domain, more attention has been drawn to generate directly spectral 3-D point clouds from HSIs and LiDAR. The most trivial way to achieve this goal is registering LiDAR with HSIs and then assigning the spectral information of HSIs to the corresponding point cloud [10]. This method is easy to perform, however it ignores the fact that HSIs are suffering from shadows, shadings, and spectral mixing and thus results in inaccurate matches. Brell *et al.* proposed a 3-D hyperspectral point cloud (HSPC) generation method that uses nonnegative matrix factorization (NMF)-based spatial unmixing to enhance the spatial resolution of HSIs and then match to the high spatial resolution point cloud [11]. However, knowledge about data acquisition and access and additional information like LiDAR range, amplitude, echo width, and inertial measurement unit (IMU)/global positioning system (GPS) information is needed to meet the requirements of the whole procedure [12], which may cause inconvenience as that information is not always available. Besides algorithms

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to generate a HSPC, efforts have also been made to develop new sensors that can acquire spectral 3-D data, such as full waveform hyperspectral LiDAR system which is based on supercontinuum laser sources [13], [14] and airborne multispectral mapping LiDAR system which uses multiwavelength LiDAR that emits laser pulses in different wavelengths [15]. The advantage of active hyperspectral imaging is that it is free of shadows and can work in the dark. What is more, the capability of producing one-shot topographic and spectral intensity information makes it free of registration problems between datasets. However, producing these kinds of systems is far more costly than using separate sensors.

The technology of integrating LiDAR instruments and hyperspectral sensors has been well established. First, powerful GPS/IMU devices can be used to tie the data streams together. Second, the airborne stability of UAVs has increased. Registration between hyperspectral and LiDAR becomes more effortless and has higher accuracy. Nevertheless, several difficulties remain to generate high-quality spectral 3-D data from coregistered HSI and LiDAR: 1) passive hyperspectral imaging is significantly influenced by environment illumination and shadows; 2) hyperspectral imaging only acquires 2-D pixels of the ground surface, which can cause mismatches of spectral and geometric information when combined with the 3-D LiDAR point cloud. To address these challenges, in this article, we present a novel supervoxel-based joint intrinsic decomposition (SJID) modal for HSIs and LiDAR. The proposed SJID model aims to recover intrinsic scene properties, such as point-wise reflectance, surface normals, and environment illumination from HSIs and LiDAR. Intrinsic scene decomposition is an inverse process of imaging, which aims to infer the 3-D physical world from 2-D images. The recovered intrinsic scene properties are thus helpful to generate spectral 3-D data faithful to the real world. Point-wise intrinsic scene decomposition of millions of points is not computational-friendly, so we further introduce a tractable supervoxel-based framework.

Typically, intrinsic image decomposition (IID) refers to the separation of reflectance and shading components from a single red-green-blue (RGB) [16]–[19]. Some more ambitious methods aim to recover at the same time shape and illumination in addition to reflectance and shading from a single image, which can be done by depth prediction using a novel convolutional neural network (CNN) [20], [21], or by natural image statistics analysis [22], the latter usually requires the inputs to be images with only single segmented objects. As for the more general case, that is, recovering the shape, illumination, and reflectance from the whole image scene where objects may occlude or support one another and spatially varying illumination exists, additional sources, either image sequences illuminated under varying illumination [23]–[25], depth cues acquired by depth sensor [26]–[29], or incident light direction [30] are needed. Though in computer vision there exists some efficient IID models for RGB or RGB-D images, these methods usually cannot be extended easily to HSIs which may contain hundreds of bands, and when they do, they usually work poorly as a remote sensing scene is much more complicated than that in computer vision. Moreover,

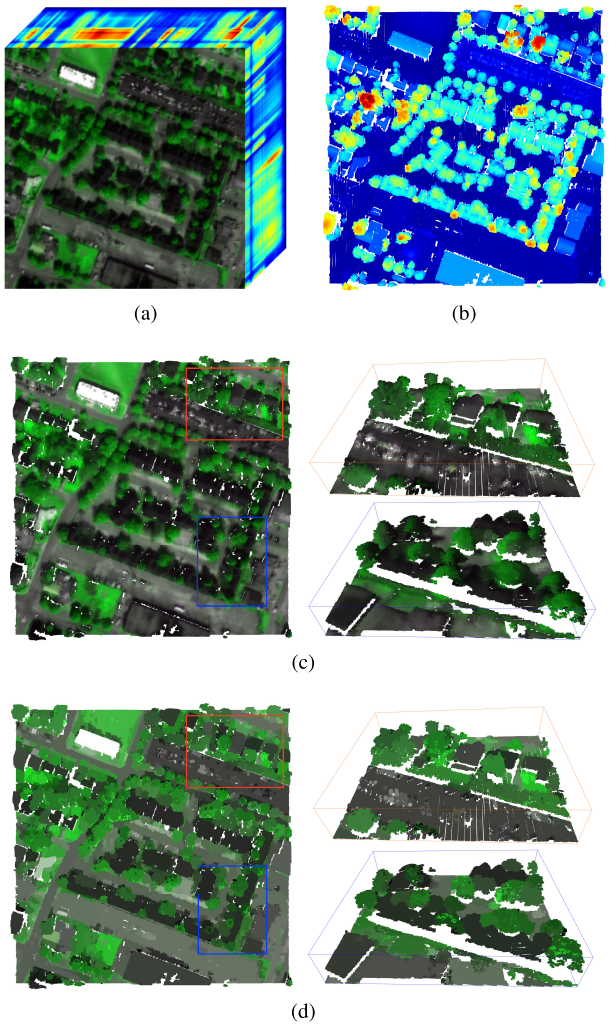

Fig. 1. Concept of IHSPC. (a) Original hyperspectral cube. (b) LiDAR point cloud. (c) HSPC via color assigning. (d) IHSPC via HSI and LiDAR joint intrinsic decomposition. The advantages of IHSPC versus HSPC are as following: 1) higher spectral homogeneous within one class, 2) reduced 3-D mismatches between spectral and spatial information, and 3) reduced shadow effects.

though the HSI "intrinsic image" problem has already been investigated by some previous works [31]–[33], researches about the more general "shape, illumination, and reflectance" problem for HSI are still lacking. In this article, we would consider the problem in the context of remote sensing image processing: given HSI and aligned LiDAR, recover intrinsic scene properties via joint IID of the two. The intrinsic HSPC (IHSPC) generated is illustrated in Fig. 1.

From the perspective of HSIs and LiDAR fusion, the difference between our work and existing methods are as follows: 1) instead of degenerating LiDAR point cloud to 2-D images that have the same spatial resolution and organized grid structure with HSI, we use directly the unorganized point clouds and explore the relation between HSI pixel and somewhat sub-pixel LiDAR points, by doing this we can make full use of the information contained in the data; 2) instead of extracting human-designed features from the two data sets and considering how to combine them, we try to physically infer the properties of a scene (coordinates, elevation, surface normals,

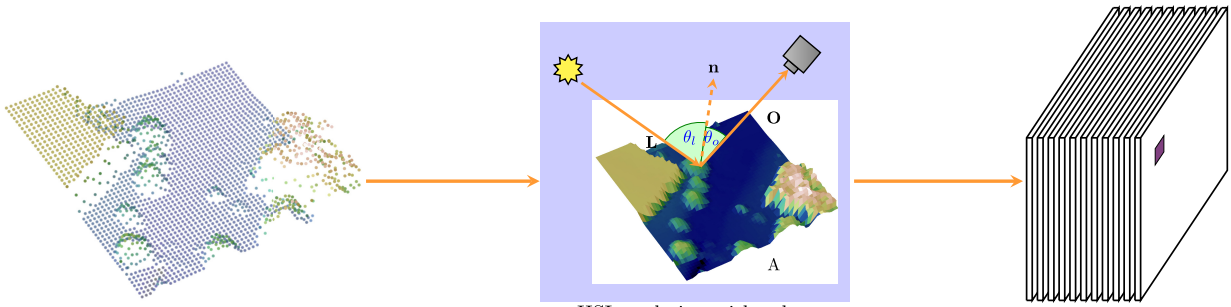

Fig. 2. Illustration of how HSI and LiDAR are connected in the proposed ISM. For each pixel of HSI, we assume some sub-pixel reflectances and shapes with render that pixel. The sub-pixel shapes are provided and estimated by discrete LiDAR points.

illumination, reflectance, etc.) that together reproduce HSIs and LiDAR. As a point-cloud-level method, optimizing with millions of remote sensing LiDAR points is rather challenging and computation expansive. To reduce the complexity of the proposed method, we further establish a supervoxel-based framework. Supervoxels are the 3-D extension of pixels. A supervoxel is a subset of similar points or voxels, which can give a meaningful and compact representation of LiDAR point cloud [34]. Supervoxel has been successfully adopted for various tasks such as labeling [35], segmentation [34], and detection [36]–[39]. To the best of our knowledge, this is the first work that combines supervoxels with HSIs and LiDAR intrinsic decomposition.

In summary, our main contributions are:

1) We proposed a computationally tractable SJID model for HSIs and LiDAR, which can recover intrinsic scene properties and finally generate an IHSPC. The key features of our approach are as follows.
    1) **Intrinsic Scene Model (ISM) for HSIs and LiDAR.** Based on bidirectional reflectance distribution function (BRDF) and Monte Carlo integration, the ISM establishes an intrinsic mapping from the 3-D point cloud to 2-D HSI pixels, which can directly handle the two data streams with different data structures and sampling resolutions. Scene properties, such as 3-D hyperspectral reflectance and environment illumination, can be effectively recovered.
    2) **A Supervoxel-Based Framework.** It is computational unfriendly to tackle millions of points or even tens of millions of points. Therefore, LiDAR point cloud is first segmented into supervoxels using a weighted-graph clustering method. Then, supervoxel-based intrinsic scene properties are recovered. Using supervoxels rather than degenerating LiDAR to 2-D images, the complexity of intrinsic decomposition tasks is greatly reduced, and at the same time, useful 3-D structural information are preserved.
2) It is challenging to quantitatively validate the effectiveness of intrinsic decomposition methods as the ground truth of natural data is hardly accessible. We extensively validate our proposed approach by using both synthetic and real datasets. We provide a range of experimental results.

The rest of this article is organized as follows. Section II describes the proposed ISM for HSI and LiDAR in detail. Section III presents how to conduct supervoxel-based intrinsic decomposition. The experimental results and analysis are reported in Section IV. Section V concludes this article.

II. ISM for HSIs and LiDAR

Our supervoxel-based model builds upon the ISM for HSIs and LiDAR proposed in our previous work [40]. The ISM addresses the challenge of combining HSIs and LiDAR that have different data structures and spatial resolutions. Since the coordinated HSIs and LiDAR are the sampling of the same ground surface, we can assume some sub-pixel shape and reflectance that render given HSI pixel and LiDAR points, as shown in Fig. 2. More specifically, we first apply BRDF to show how each pixel in an HSI is rendered with sub-pixel reflectances and shapes in the continuous domain, then we apply Monte Carlo integration to derive the discrete form of the proposed model and thus connect HSIs with the unorganized LiDAR point cloud. If HSI pixel $\mathbf{h}_i$ and LiDAR points $\{\mathbf{p}_j\}_{j=1}^k$ belong to same ground surface A, the ISM model says $\mathbf{h}_i$ can be rendered from $\{\mathbf{p}_j\}_{j=1}^k$ by

$$\mathbf{h}_i(\lambda) = E_l \frac{\sum_{j=1}^k \mathbf{r}_j(\lambda)(\mathbf{L}^\intercal \mathbf{n}_j)(\mathbf{O}^\intercal \mathbf{n}_j)}{\sum_{j=1}^k (\mathbf{O}^\intercal \mathbf{n}_j)} \tag{1}$$

where λ is the wavelength, $\mathbf{r}_j$ is the reflectance of point $\mathbf{p}_j$, ω_i is the incident light direction, ω_r is the observation direction of the sensor, $\mathbf{n}_j$ is the surface normal on $\mathbf{p}_j$, and E_l is the radiant intensity of the incident light. For more details, we refer the readers to [40].

The advantage of ISM is that it directly respects how images are physically formed by utilizing BRDF and geometry-based Monte-Carlo sampling. The intrinsic decomposition problem can be accurately solved by perusing the inverse of it. However, the calculation can be rather expensive when the scene is large (it is usually true for remote sensing images). This critically limited the application of the ISM model. To address

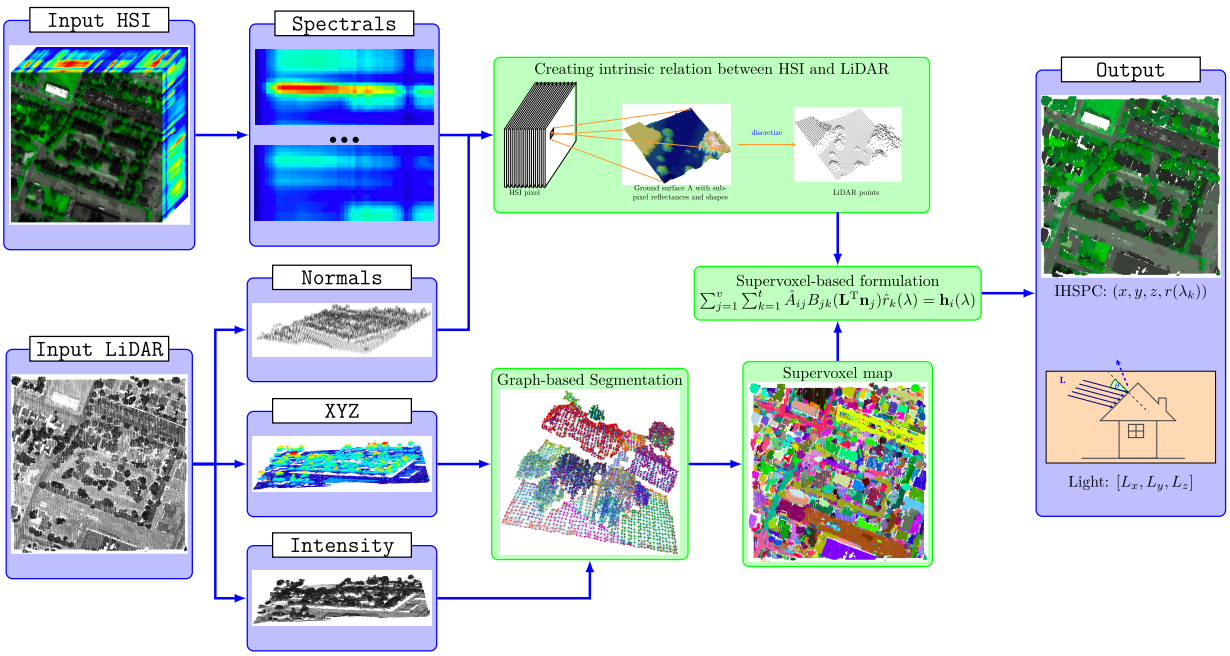

Fig. 3. Flowchart of the proposed method.

this shortcoming, we further proposed a supervoxel-based ISM which can still preserve the main properties of ISM as well as greatly reduce the complexity.

III. Supervoxel-Based Intrinsic Scene Decomposition

In this section, we will discuss our supervoxel-based intrinsic scene decomposition method that is based on (1) in detail. The general flowchart of the proposed method is given in Fig. 3.

A. Point-Cloud-Level Formulation

Equation 1 gives the rendering formulation for a given HSI pixel and LiDAR points, and it can be inconvenient to apply to the whole scene. Therefore, we here try to describe the intrinsic scene decomposition problem for HSIs and LiDAR in a more general form: given HSI, $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \ldots, \mathbf{h}_u]^{\mathsf{T}} \in \mathbb{R}^{u\times d}$ and LiDAR point cloud, $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \ldots, \mathbf{p}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times 4}$, recover point-wise reflectance $\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times d}$, and incident light direction $\mathbf{L} = [L_x, L_y, L_z]^{\mathsf{T}} \in \mathbb{R}^3$, where u is the number of HSI pixels, v is the number of LiDAR points, $\mathbf{h}_i = [h_i(\lambda_1), h_i(\lambda_2), \ldots, h_i(\lambda_d)]^{\mathsf{T}}$ is the d-dimensional spectral measure for pixel i, $\mathbf{p}_i = [x_i, y_i, z_i, I_i]^{\mathsf{T}}$ is the coordinates and intensity of LiDAR point i, and $\mathbf{r}_i = [r_i(\lambda_1), r_i(\lambda_2), \ldots, r_i(\lambda_d)]^{\mathsf{T}}$ is the reflectance for point i. For simplicity, in this work, we ignore the outgoing light direction estimate problem and just set $\mathbf{O}$ as $[0, 0, 1]^{\mathsf{T}}$. Moreover, surface normals are denoted as $\mathbf{N} = [\mathbf{n}_1, \mathbf{n}_2, \ldots, \mathbf{n}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times 3}$, where $\mathbf{n}_i = [N_i^x, N_i^y, N_i^z]^{\mathsf{T}} \in \mathbb{R}^3$ is the normal direction on point i. The symbols and definitions are summarized in Table I.

We further define a matrix $\mathbf{A} \in \mathbb{R}^{u\times v}$ which indicates whether certain LiDAR points fall into an HSI pixel:

$$A_{ij} = \begin{cases} 1, & \text{if } \mathbf{p}_j \text{ falls in } \mathbf{h}_i \\ 0, & \text{ortherwise.} \end{cases} \tag{2}$$

TABLE I
Symbols and Definitions

Symbol	Definition
$\mathbf{H}$	Input HSI, $\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \cdots, \mathbf{h}_u]^{\mathsf{T}} \in \mathbb{R}^{u\times d}$.
$\mathbf{h}_i$	Spectral of pixel i, $\mathbf{h}_i = [h_i(\lambda_1), h_i(\lambda_2), \cdots, h_i(\lambda_d)]^{\mathsf{T}}$.
$\mathbf{P}$	LiDAR point cloud, $\mathbf{P} = [\mathbf{p}_1, \mathbf{p}_2, \cdots, \mathbf{p}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times 4}$.
$\mathbf{p}_i$	Coordinates and intensity of point k, $\mathbf{p}_i = [x_i, y_i, z_i, I_i]^{\mathsf{T}}$.
$\mathbf{N}$	Surface normals, $\mathbf{N} = [\mathbf{n}_1, \mathbf{n}_2, \cdots, \mathbf{n}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times 3}$.
$\mathbf{n}_i$	Surface normal on point i, $\mathbf{n}_i = [N_i^x, N_i^y, N_i^z]^{\mathsf{T}} \in \mathbb{R}^3$.
$\mathbf{L}$	Incident light direction, which is denoted as $[L_x, L_y, L_z]^{\mathsf{T}} \in \mathbb{R}^3$ in Cartesian system.
$\mathbf{O}$	Observed direction of the sensor, we ignore the estimation problem of $\mathbf{O}$ in this paper and set it as $[0, 0, 1]^{\mathsf{T}}$.
$\mathbf{R}$	Point-wise hyperspectral reflectance, $\mathbf{R} = [\mathbf{r}_1, \mathbf{r}_2, \cdots, \mathbf{r}_v]^{\mathsf{T}} \in \mathbb{R}^{v\times d}$.
$\hat{\mathbf{R}}$	Supervoxel-based reflectance, $\hat{\mathbf{R}} = [\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \cdots, \hat{\mathbf{r}}_v]^{\mathsf{T}} \in \mathbb{R}^{t\times d}$.

As we set $\mathbf{O}$ as $[0, 0, 1]^{\mathsf{T}}$, then $\mathbf{O}^{\mathsf{T}}\mathbf{n}_j = N_j^z$. In (1), we need to specify which subset $\{\mathbf{p}_j\}_{j=1}^k$ fall in pixel i, we use (2) to achieve this purpose and rewrite (1) as

$$\begin{aligned} \mathbf{h}_i(\lambda) &= E_l \frac{\sum_{j=1}^{v} A_{ij} N_j^z \mathbf{r}_j(\lambda)(\mathbf{L}^{\mathsf{T}}\mathbf{n}_j)}{\sum_{j=1}^{v} A_{ij} N_j^z} \\ &= \sum_{j=1}^{v} \hat{A}_{ij} \mathbf{r}_j(\lambda)(\mathbf{L}^{\mathsf{T}}\mathbf{n}_j) \end{aligned} \tag{3}$$

where $\hat{A}_{ij} = E_l A_{ij} N_j^z / (\sum_{j=1}^{v} A_{ij} N_j^z)$.

B. LiDAR Supervoxel Segmentation

Given LiDAR point cloud $\mathbf{P}$, a supervoxel oversegmentation is a partition $S = \{S_i\}_{i=1}^t$ of $\mathbf{P}$ such that $\mathbf{P} = \cup_i^t S_i$ and $S_i \cap S_j = \emptyset$, if $i \neq j$, where t is the number of supervoxels. In this work, we transform the supervoxel segmentation problem into weighted graph clustering problem to get spatially and spectrally homogeneous segments. Initially,

t is set to v which is same as the number of LiDAR points; in this case, each supervoxel S_k is a LiDAR point k itself. Hence, we create a weighted graph $G = (V, E)$ and initialize G in the following way: 1) each node $v_i \in V$ denotes a LiDAR point $\mathbf{p_i}$; 2) node i and node j are joined by an edge if $||(x_i, y_i, z_i) - (x_j, y_j, z_j)|| < \delta_x$, where $||(x_i, y_i, z_i) - (x_j, y_j, z_j)||$ calculates the Euclidean distance between two LiDAR points; δ_x is the radius specified by the user, only neighbors within this radius of a given point are linked to it with an edge.; 3) the weight $W(e)$ of edge joining node i and j is measured as $||I_i - I_j||$, where I_i and I_j are the intensity of $\mathbf{p}_i$ and $\mathbf{p}_i$, respectively, $||I_i - I_j||$ measures the dissimilarity between $\mathbf{p}_i$ and $\mathbf{p}_i$. Then, our supervoxel clustering strategies are as following: 4) rank all edges in V from small to large by weight; 5) for each edge e in the ranking, if the two nodes i and j joined by e belong to different supervoxels S_a and S_b, then determine if $W(e) < \mathrm{Dis}(S_a, S_b) + \delta_I$, if so, merge S_a and S_b, where $\mathrm{Dis}(S_a, S_b)$ measures the dissimilarity between supervoxels S_a and S_b, it is defined as the smallest weight of the edges that link one point is S_a and the other point in S_b; δ_I is a threshold. The supervoxel method is summarized in algorithm 1.

Algorithm 1 Supervoxel Segmentation

1: Given the input LiDAR point cloud $\mathbf{P}$, initialize supervoxel $\{S_i\}_{i=1}^{v}$ and weighted graph $G = (V, E)$
2: Rank the edge $e \in E$ by weights from small to big
3: **for all** $e \in E$ **do**
4: **if** the two nodes V_i and V_j joined by e belong to different supervoxels S_a and S_b **then** go to 5
5: **if** $W(e) < Dis(S_a, S_b) + \delta_I$ **then** Merge supervoxels S_a and S_b
6: **end if**
7: **else** Return to 3
8: **end if**
9: **end for**
10: Output the updated supervoxel set $\{S_i\}_{i=1}^{t}$

C. Supervoxel-Based Optimization

For every supervoxel, since it is a spatially and spectrally homogeneous segment, we think every point in it has same reflectance. Hence, we define the supervoxel-based reflectance as $\hat{\mathbf{R}} = [\hat{\mathbf{r}}_1, \hat{\mathbf{r}}_2, \ldots, \hat{\mathbf{r}}_t]^\intercal \in \mathbb{R}^{t \times d}$.

For every HSI pixel pair $\mathbf{h}_a$ and $\mathbf{h}_b$, if they fall entirely in one supervoxel, we can get an equation set as

$$\begin{cases} \mathbf{L}^\intercal \left(\sum_{j=1}^{v} \hat{A}_{aj} \mathbf{n}_j \right) \hat{r}(\lambda_k) = h_a(\lambda_k) \\ \mathbf{L}^\intercal \left(\sum_{j=1}^{v} \hat{A}_{bj} \mathbf{n}_j \right) \hat{r}(\lambda_k) = h_b(\lambda_k) \end{cases} \tag{4}$$

$$k = 1, 2, \ldots, d. \tag{5}$$

Furthermore, we have

$$\frac{1}{\hat{r}(\lambda_k)} = \mathbf{L}^\intercal \sum_{j=1}^{v} \frac{\hat{A}_{aj}}{h_a(\lambda_k)} \mathbf{n}_j = \mathbf{L}^\intercal \sum_{j=1}^{v} \frac{\hat{A}_{bj}}{h_b(\lambda_k)} \mathbf{n}_j. \tag{6}$$

Therefore, $\hat{r}(\lambda_k)$ can be eliminated and over-determined equations about $\mathbf{L}$ can be derived

$$\mathbf{L}^\intercal \sum_{j=1}^{v} \left(\left(\frac{\hat{A}_{aj}}{h_a(\lambda_k)} - \frac{\hat{A}_{bj}}{h_b(\lambda_k)} \right) \mathbf{n}_j \right) = 0. \tag{7}$$

Solving the over-determined equations systems given by (7), $\mathbf{L}$ can be estimated.

Since $\mathbf{L}$ is estimated, we can then use $\mathbf{L}$ to calculate $\hat{\mathbf{R}}$. Similar to $\mathbf{A}$, we define a matrix $\mathbf{B} \in \mathbb{R}^{v \times t}$ which describes the mapping from $\mathbf{P}$ to S such that

$$B_{ij} = \begin{cases} 1, \text{if } \mathbf{p}_i \in S_j \\ 0, \text{ ortherwise.} \end{cases} \tag{8}$$

Using $\mathbf{B}$, we can map $\hat{\mathbf{R}}$ to $\mathbf{R}$ by

$$r_j(\lambda) = \sum_{k=1}^{t} B_{jk} r_{sk}(\lambda). \tag{9}$$

Substituting (9) to (3), we get

$$\begin{aligned} \mathbf{h}_i(\lambda) &= \sum_{j=1}^{v} \sum_{k=1}^{t} \hat{A}_{ij} B_{jk} (\mathbf{L}^\intercal \mathbf{n}_j) \hat{r}_k(\lambda) \\ &= \sum_{j=1}^{v} \sum_{k=1}^{t} \hat{A}_{ij} \hat{B}_{jk} \hat{r}_k(\lambda) \end{aligned} \tag{10}$$

where $\hat{B}_{jk} = B_{jk}(\mathbf{L}^\intercal \mathbf{n}_j)$.

The matrix form of (10) is

$$\mathbf{M}\hat{\mathbf{R}} = \mathbf{H} \tag{11}$$

where $\mathbf{M} = \hat{\mathbf{A}}\hat{\mathbf{B}} \in \mathbb{R}^{u \times t}$.

In order to solve the underdetermined equation system (11), we further propose a constraint matrix $\mathbf{W}$ that make $\hat{\mathbf{R}}$ has similar reflectance locally. $\mathbf{W}$ is defined as

$$\hat{\mathbf{r}}_i = \sum_{j=1}^{t} W_{ij} \hat{\mathbf{r}}_j \tag{12}$$

where L1-graph coding is used to calculate W_{ij} [41]

$$\min_{\alpha_i} \|\alpha_i\|_1 \quad \text{s.t. } \mathcal{I}_i = \hat{\mathbf{h}}_i \alpha_i \tag{13}$$

where matrix $\mathbf{D}^i = \left[\hat{\mathbf{h}}_1, \ldots, \hat{\mathbf{h}}_{i-1}, \hat{\mathbf{h}}_{i+1}, \ldots, \hat{\mathbf{h}}_t, \mathbf{I}_d\right]$, $\mathbf{I}_d$ is a $d \times d$ unit matrix; $\alpha_i \in R^{d+t-1}$ is the sparse coefficients; $\hat{\mathbf{h}}_i$ is the mean spectra of supervoxel i. Then, set $W_{ij} = \alpha_i(j)$ if $i > j$, and $W_{ij} = \alpha_i(j-1)$ if $i < j$.

Finally, the constrained cost function is given as

$$U = \left\| \mathbf{M}\hat{\mathbf{R}} - \mathbf{H} \right\|_{\mathrm{F}}^2 + \delta_u \left\| \mathbf{W}\hat{\mathbf{R}} - \hat{\mathbf{R}} \right\|_{\mathrm{F}}^2 \tag{14}$$

where δ_u is the tradeoff between the two terms. Compute the first-order partial derivative, we have

$$\frac{1}{2} \frac{\partial U}{\partial \hat{\mathbf{R}}} = [\mathbf{M}^\intercal \mathbf{M} + \delta_u (\mathbf{W}^\intercal - \mathbf{I}_t)(\mathbf{W} - \mathbf{I}_t)] \hat{\mathbf{R}} - \mathbf{M}^\intercal \mathbf{H}. \tag{15}$$

Solve the linear system $\partial U / \partial \hat{\mathbf{R}} = 0$, we then get $\hat{\mathbf{R}}$. Solving (14) we can recover reflectance as $\mathbf{R} = \mathbf{B}\hat{\mathbf{R}}$. By utilizing supervoxel, the complexity of optimization reduced greatly from $v \log(v)$ to $t \log(t)$, since $t \ll v$.

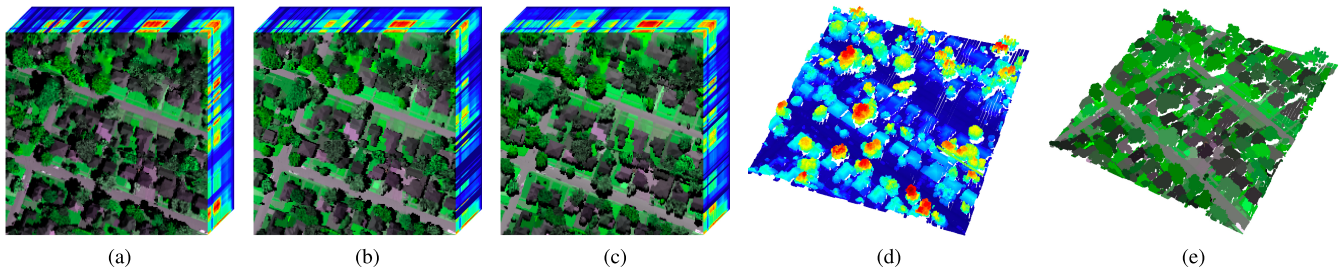

Fig. 4. Pseudo-synthetic HSIs and LiDAR with ground truth HSPC. HSI_light1 (a), HSI_light2 (b), and HS_light3 (c) are the three synthetic HSIs of different light conditions. (d) Height ramp of LiDAR point cloud. (e) Ground-truth IHSPC.

IV. Experimental Results and Discussion

In this section, the results of the experiments on both synthetic and real data are reported to evaluate the effectiveness of the proposed SJID model for HSIs and LiDAR. The visualization of the 3-D point cloud and some functions of our method use the OPEN3D library [42], which is an open-source library that supports rapid development of software that deals with 3-D data.

A. Experiments on Pseudo-Synthetic Data

1) Data Description: One challenge we meet is that it is rather difficult to produce ground-truth hyperspectral reflectance, illumination, and normals for real natural scenes. To quantitatively evaluate the performance of the proposed method, we compose pseudo-synthetic scenes based on the "grss_dfc_2018" dataset, which is used for the 2018 IEEE Geoscience and Remote Sensing Society (GRSS) Data Fusion Contest [43]. Our idea is first to generate an IHSPC which we can deem as ground truth and based on which we can synthesize a series of HSIs on different illuminations. More specifically, we take a sub-scene from "grss_dfc_2018" dataset with LiDAR point cloud and corresponding HSI. We first run the hyperspectral image decomposition algorithm in [32] on HSI to obtain a 2-D reflectance image, which still suffers from spectral mixing and shadow effects. Then, we manually segment the point cloud into different objects and manually assign reflectance from the 2-D reflectance image for each object (avoiding using reflectance from image boundaries and shadow regions). With the 3-D terrain, 3-D surface reflectances from the generated IHSPC, we then synthesize three HSIs with different illuminations with a discrete Monte Carlo approach. The data is illustrated in Fig. 4, which includes five sub-data: three HSI cubes synthesized with different light conditions, the LiDAR point cloud, the ground-truth IHSPC of the scene.

2) Results and Analysis: We apply the proposed SJID method on the three HSIs of the different light conditions and the LiDAR point cloud. The alternative methods we compared are: 1) directly assigning the original spectral information of HSI to aligned LIDAR point cloud and thus generating a HSPC, which is a base-line method and denoted by "HSPC_ori"; 2) first conducting hyperspectral image decomposition algorithm in [32] on the HSI and then assigning the obtained 2-D hyperspectral reflectance to the aligned LiDAR point cloud, we denote this as "HSPC_2d_ref"; 3) generating IHSPC using the proposed joint supervoxel-based intrinsic decomposition method for HSI and LiDAR. The visual comparison of the generated IHSPCs with alternative methods is shown in Fig. 5, from which we can see that: 1) HSPC_ori suffers from both shading and shadow effects, the light condition has a great impact on the generated HSPC, surface normal changes will also cause spectrum changes, which is not expected. Moreover, due to the small misalignments between the HSI and the LiDAR, there are mismatches of spectral and spatial information in edge regions. 2) HSPC_2d_ref can reduce spectrum changes caused by surface normal changes; however, it is still influenced by shadows and misalignments. 3) The proposed SJID method can generate an IHSPC that is free of light conditions. Moreover, by creating intrinsic mapping relation from the LiDAR to the HSI it can recover sub-pixel reflectances and thus is robust to small misalignments.

The quantitative R-mean squared error (MSE) of different methods are given Fig. 6, the proposed method has the smallest error and it is not sensitive to light conditions. The R-MSE error for a 3-D IHSPC is computed as

$$\mathrm{R}_{\mathrm{MSE}}(\hat{\mathbf{R}}, \mathbf{R}^*) = \frac{1}{v} \sum_{\lambda} \min_{\alpha} \sum_{x,y,z} \left\| \alpha \hat{\mathbf{R}}_{x,y,z}(\lambda) - \mathbf{R}^*_{x,y,z}(\lambda) \right\|_2^2 \quad (16)$$

where $\mathbf{R}^*$ is the ground-truth hyperspectral reflectance and $\hat{\mathbf{R}}$ is the recovered hyperspectral reflectance.

The proposed method also gives an estimation of the light direction of the HSI scene. Table II shows the estimated light direction and the error compared with ground-truth light direction, where the error is measured by

$$\mathrm{L}_{\mathrm{err}}(\hat{\mathbf{L}}, \mathbf{L}^*) = \arccos\left(< \hat{\mathbf{L}}, \mathbf{L}^* >\right). \quad (17)$$

3) Influence of Parameters: In this section, we analyze the impact of different parameter settings on R-MSE. There are three user-defined parameters for our method, that is, δ_x, δ_I, and δ_u. δ_x and δ_I affect the supervoxel segmentation result. We set δ_x as the double of the nominal spatial resolution of LiDAR point cloud, then one point is linked to about 10–20 neighboring points. δ_I is a more important parameter, which determines the scale of the supervoxels. δ_I is empirically determined using a heuristic method. More specifically, we can input a subset of the LiDAR point cloud to try out some preferred parameter settings. We report the changes in the number of supervoxels and R-MSE with the increase of δ_I in Fig. 7. We can find that the larger δ_I is, the smaller the

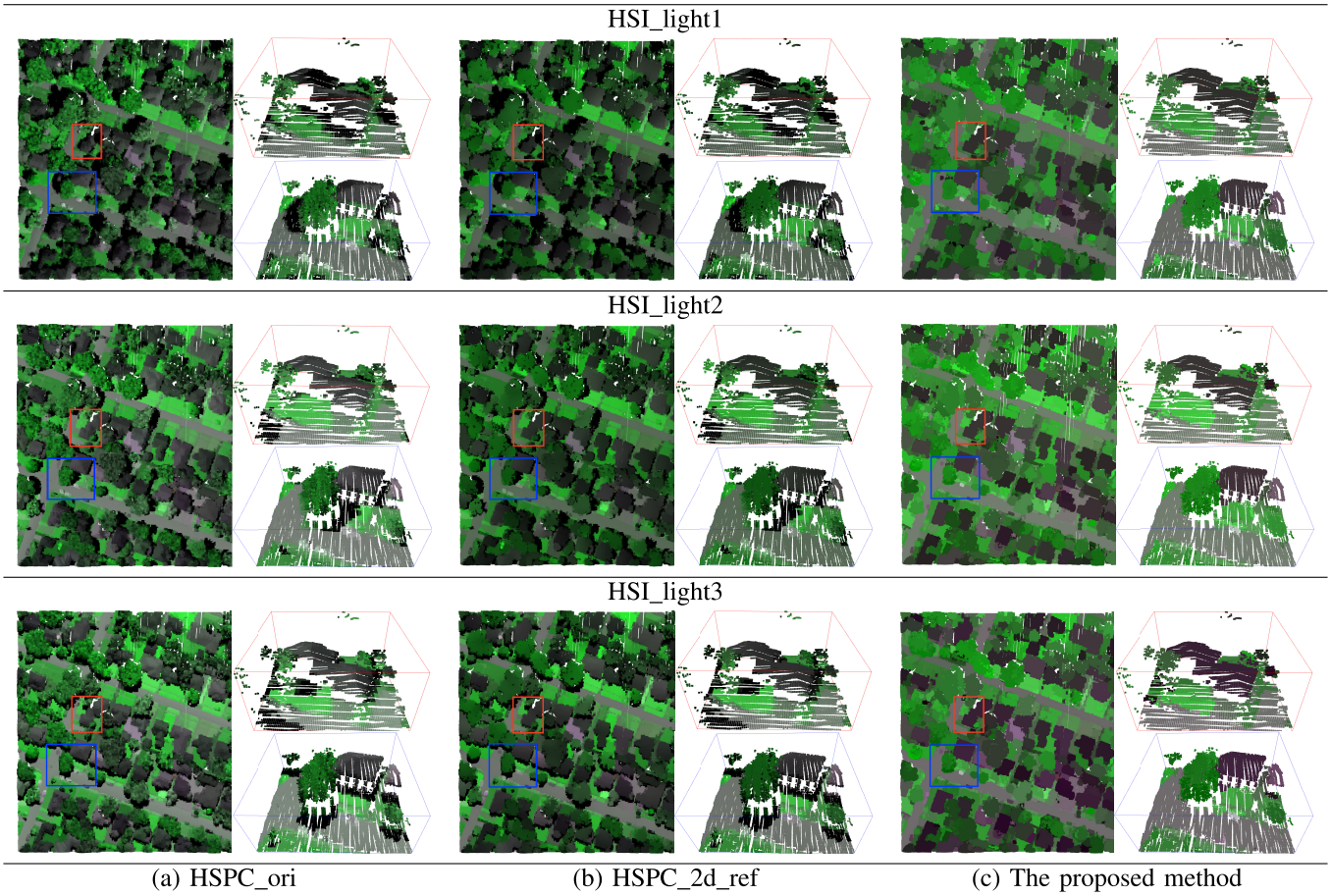

Fig. 5. HSPC generated from the three HSIs and LiDAR by different methods. (a) HSPC generated by assigning original spectral information to LiDAR point cloud. (b) HSPC generated by assigning the 2-D reflectance of HSI to LiDAR point cloud. (c) IHSPC generated by the proposed SJID method.

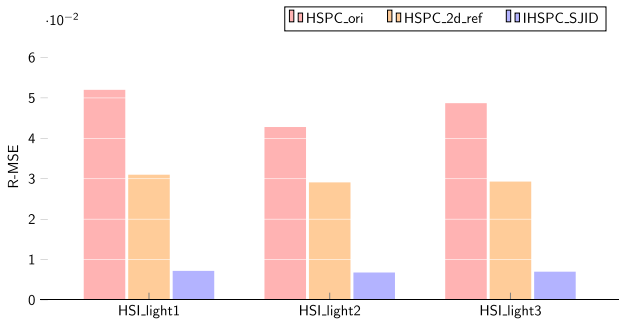

Fig. 6. R-MSE of different methods on the three HSI datasets.

number of supervoxels would be (see the blue line). We can also find that R-MSE is not influenced greatly by δ_I in the interval $[0.001, 0.1]$, which demonstrates the robustness of our method. The R-MSE is much bigger when $\delta_I = 1$, and this is because a too large δ_I would introduce over-segmentation errors. We set $\delta_I = 0.05$ for all data sets. Fig. 8 illustrates the impact of δ_u. The best δ_u varies with the data. For scene HSI_light1, he best δ_u is 0.5, while for HSI_light2 and HSI_light3, the best δ_u is near 10. Overall, the variance is not very significant.

TABLE II
ESTIMATED LIGHT DIRECTION ON THE 3 DATA

Data	Ground-truth	Estimated	Error
HSI_light1	$[0.5, -0.5, 0.71]$	$[0.40, -0.43, 0.81]$	0.158
HSI_light2	$[-0.5, 0, 0.87]$	$[-0.46, -0.07, 0.89]$	0.083
HSI_light3	$[0, 0.5, 0.87]$	$[-0.11, 0.46, 0.88]$	0.117

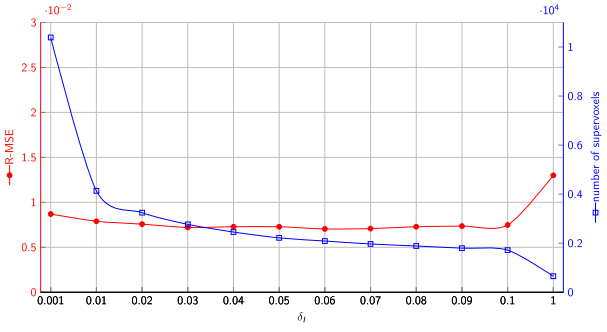

Fig. 7. R-MSE with the increase of δ_I on data HSI_light1.

4) Computing Time: We report the computing time in Table III. The tests are run using an Intel Core i7-6800K processor and 128 GB of RAM. We can see that using

TABLE III
COMPUTING TIME OF THE PROPOSED METHOD

Scene	Image size	Point size	Without supervoxel	With supervoxel
HSI_light1	40000	919760	$38m21s$	$1m12s$
HSI_light2	40000	919760	$39m12s$	$1m09s$
HSI_light3	40000	919760	$39m41s$	$1m17s$

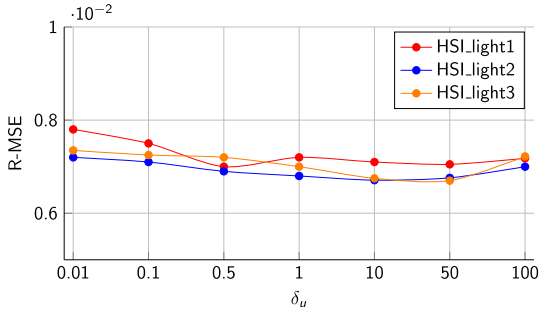

Fig. 8. R-MSE with the increase of δ_u.

supervoxel representation, the computing time of HSI and LiDAR intrinsic decomposition is dramatically reduced. It widens the application of our method in large remote sensing scenes.

B. Experiments on Real Data

1) Data Description: Another possible way of evaluation is to use a downstream task-like point cloud classification. The intrinsic hyperspectral reflectance could eliminate the impact of illumination and shading, which would follow an increase in material discrimination ability. The data we use is scene 272056_3289689 of "grss_dfc_2018" dataset. The "grss_dfc_2018" dataset was collected by a single flight conducted on February 16, 2017, which was meant for multisource optical remote sensing data fusion and analysis, and provide the following multisource data acquired by three different sensors:

1) Multispectral LiDAR point cloud data, intensity rasters, and DSMs at a 0.5-m ground sampling distance (GSD), recording of three spectral channels (1 550, 1 064, 532 nm).
2) Hyperspectral data at a 1-m GSD, spectral range 380–1 050 nm, recording of 48 spectral bands.
3) Very high-resolution (VHR) RGB imagery at a 5-cm. GSD

The scene 272056_3289689 we use including an HSI with a resolution of 601 × 596, 48 spectral bands, a multispectral point cloud (MSI) with three spectral channels, and a very-high-resolution RGB image. As for the MSI, we only use the intensity information of channel 1 064 nm and the outliers of the point cloud are removed. We make a 3-D annotation of the point cloud with the assistance of very-high-resolution RGB images of the scene, and finally, ten classes are given, as shown in Fig. 9.

As for 3-D classification, the 0.5% of the labeled samples were randomly selected as training samples and the rest as test samples; more details are given in Table IV.

TABLE IV
TRAINING AND TESTING SAMPLES FOR EACH CLASS

No.	Scene 272056_3289689		
	Categories	Training Samples	Test Samples
C1	Evergreen trees	3945	784922
C2	Deciduous trees	6603	1313885
C3	Grass	8273	1646143
C4	Non-residential buildings	3414	679351
C5	Residential buildings	3680	732241
C6	Road	7650	1522158
C7	Parking lot	4249	845520
C8	Bare earth	1030	204823
C9	Other grounds	9620	1910658
C10	Car	1215	241715
	Total	49661	9881416

2) Results and Analysis: To demonstrate the classification ability of the generated IHSPC, we conduct a comparison experiments using support vector machine (SVM) with eight different feature inputs:

1) MSPC: Three spectral intensity features from the three-channel multispectral LiDAR point cloud.
2) HSPC_ori: 48 spectral features from the HSPC generated by assigning original spectral information to point cloud.
3) HSPC_2d_ref: 48 spectral features from the HSPC generated by assigning 2-D reflectance recovered by [32] to the point cloud.
4) IHSPC_SJID: 48 spectral features from IHSPC proposed SJID of the HSI and the point cloud.
5) MSPC+Z: Three spectral intensity features from the MSPC + elevation feature of point cloud.
6) HSPC_ori+Z: 48 spectral features from the HSPC + elevation feature of point cloud.
7) HSPC_2d_ref+Z: 48 spectral features from the HSPC_2d_ref + elevation feature of point cloud.
8) DGF: features extracted by discriminative graph-based fusion (DGF) method [3]. Note, DGF extracts 2-D features; we have mapped them to 3-D LiDAR point cloud for classification.
9) IHSPC_SJID+Z: 48 spectral features from IHSPC + elevation feature of point cloud.
10) IHSPC_SJID+I: 48 spectral features from IHSPC + intensity feature of point cloud.
11) IHSPC_SJID+N: 48 spectral features from IHSPC + 3 normal features of point cloud.

The parameter settings of SVM are as following: the radial basis function (RBF) kernel scale was fixed to 10, the penalization parameter C was set to 1000 for all experiments.

Fig. 9. Scene 272056_3289689 of "grss_dfc_2018" dataset. (a) HSI. (b) LiDAR point cloud. (c) Ground-truth classification map. (d) Legend.

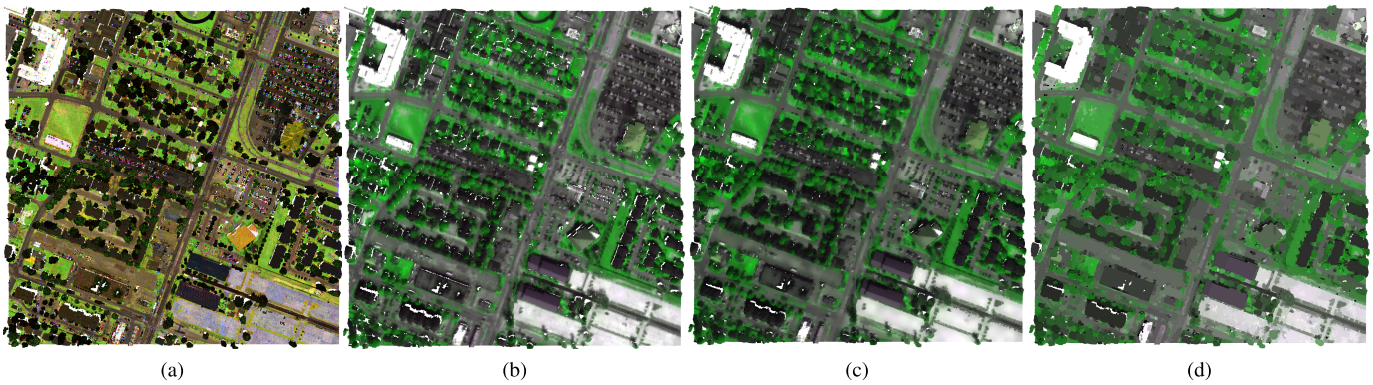

Fig. 10. Colored point cloud visualization of for kind of data. (a) MSPC. (b) HSPC_ori. (c) HSPC_2d_ref. (d) IHSPC_SJID.

Under these parameters, the training accuracy of each feature is around 92%.

The colored point cloud visualizations of the MSPC, HSPC_ori, HSPC_2d_ref, and IHSPC_SJID are given in Fig. 10. We can see that the MSPC is not influenced very much by illumination and shadows, since it is active multispectral imaging. Moreover, it does not have to worry about the registration problem between geometric and spectral information. Still, the intensities of MSPC are influenced by the scanning angles of the laser and are suffered from specular reflection on smooth surfaces. As a result, its spectral information seems a bit noisy. Another drawback is that it only has three spectral channels and is difficult to get more due to hardware constraints. The HSPC_ori, on the other hand, is influenced greatly by illumination, shadows, and misalignments between the HSI and the LiDAR. HSPC_2d_ref can reduce the illumination influence to a certain extent, but other impacts still remain. In contrast, IHSPC_SJID generated by the proposed methods can effectively reduce the illumination and shadow impacts and at the same time handle the misalignment problem.

Table V reports the classification accuracy over each class, the average accuracy (AA), and the overall accuracy (OA) of various features on Scene 272056_3289689. The previous four methods use only spectral information, while the latter four methods use both spectral and geometric information. When only spectral information is used, we can see that the MSPC feature results in the worst classification performance, which shows that, in general, multispectral feature has weaker land cover material classification ability than hyperspectral features. HSPC_2d_ref achieves better classification results than HSPC_ori in every class, which means 2-D hyperspectral reflectance does make a change, but the improvement is limited. The 3-D hyperspectral reflectance from IHSPC_SJID in contrast can improve the accuracy of every class over 8% and in total improve the OA and AA over 10%. From the classification confusion matrix given in Fig. 11, we can see more clearly where the improvement is come from: while MSPC, HSPC_ori, and HSPC_2d_ref will mistakenly classify grass, trees, and buildings with each other, the proposed IHSPC_SJID can accurately classify them. We can observe that the first four features all perform poorly on classes like bare earth and cars. The reasons may be that these two classes have a relatively much smaller number of samples and small objects like cars are more easily influenced by noises and misalignments. Comparing the results of MSPC+Z, and HSPC_ori+Z, HSPC_2d_ref+Z, and IHSPC_SJID+Z compared with those of MSPC, and HSPC_ori, HSPC_2d_ref, and IHSPC_SJID, we can see that combining spectral information with geometric information like elevation can bring an improvement in classification accuracy, especially for classes like trees, buildings, and cars. Note that the proposed IHSPC_SJID which uses only spectral information already beats MSPC+Z, HSPC_ori+Z, and HSPC_2d_ref+Z

TABLE V
CLASSIFICATION ACCURACY (%) OF VARIOUS FEATURES ON THE SCENE 272056_3289689

Methods	Evergreen trees	Deciduous trees	Grass	Non-residential buildings	Residential buildings	Road	Parking lots	Bare Earth	Other grounds	Car	AA	OA
MSPC	39.31	93.72	84.57	31.32	82.08	71.06	2.25	0.05	74.25	22.57	50.12	63.95
HSPC_ori	75.45	77.96	71.34	87.56	77.83	76.66	64.00	46.87	74.68	39.73	69.21	73.70
HSPC_2d_ref	77.74	80.76	71.72	90.20	81.58	83.54	72.73	55.38	78.23	39.63	73.15	77.44
IHSPC_SJID	91.29	**96.66**	85.18	98.85	98.14	**94.46**	83.25	**66.48**	86.26	54.04	85.46	89.42
MSPC+Z	40.74	93.55	85.10	81.71	85.55	73.83	31.52	0.06	75.22	92.64	65.99	72.69
HSPC_ori+Z	81.34	87.86	83.16	94.67	88.55	78.96	72.40	51.24	76.94	85.57	80.07	81.46
HSPC_2d_ref+Z	82.62	88.95	83.27	95.24	89.65	85.62	80.30	59.63	80.57	86.00	83.19	84.43
DGF	82.58	90.86	85.82	93.82	91.85	84.34	74.95	56.00	76.44	86.66	82.33	83.66
IHSPC_SJID+Z	**92.12**	95.49	**85.88**	**99.36**	**98.52**	94.11	**85.36**	65.74	**86.44**	**93.22**	**89.62**	**90.61**
IHSPC_SJID+I	91.36	95.39	87.69	98.51	97.92	91.60	80.43	56.41	83.76	63.51	84.66	88.49
IHSPC_SJID+N	84.43	89.77	85.76	97.28	96.88	91.67	79.09	55.31	83.07	59.59	82.29	86.35

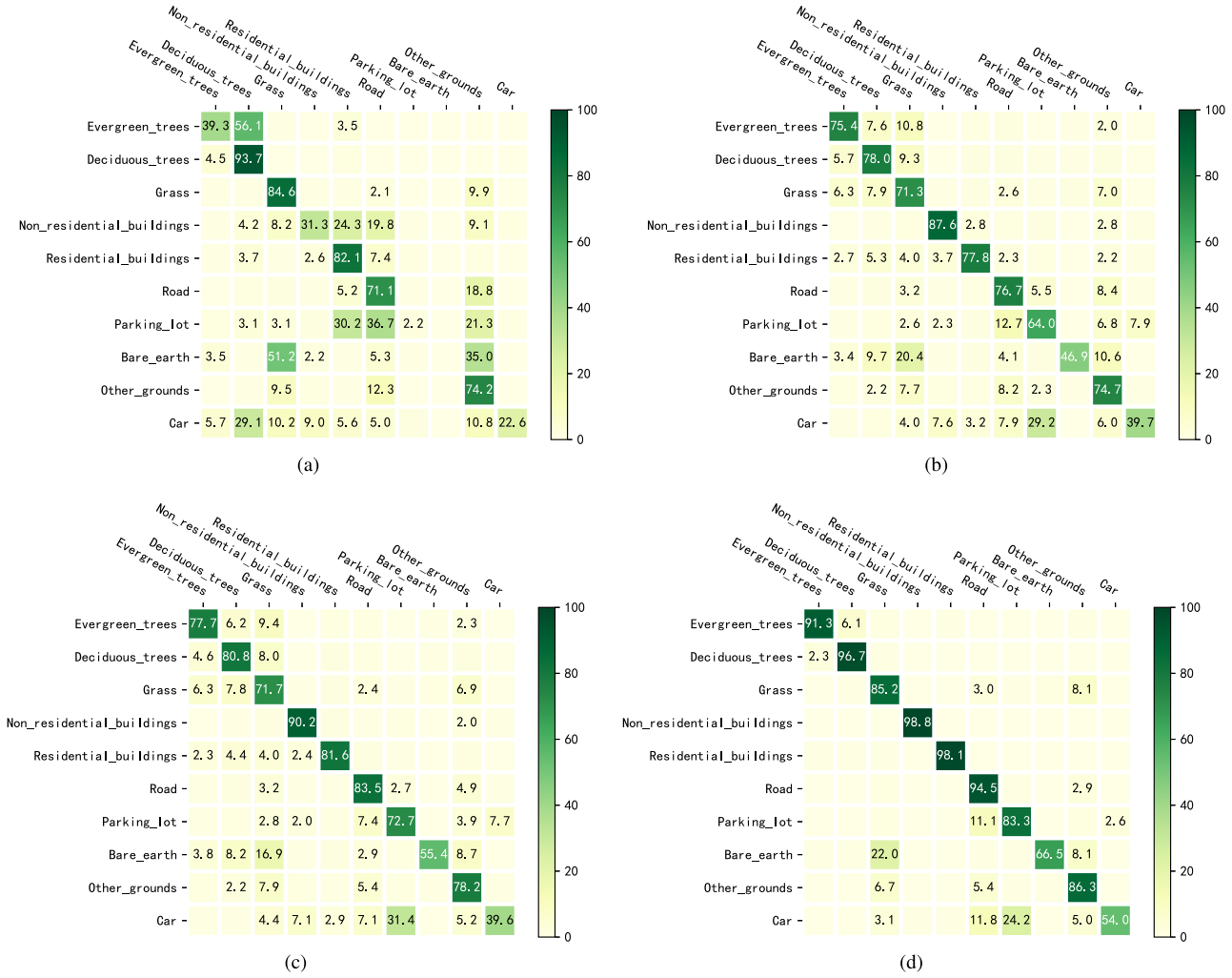

Fig. 11. Confusion matrix of different methods. (a) MSPC. (b) HSPC_ori. (c) HSPC_2d_ref. (d) IHSPC_SJID.

that use both spectral and elevation information, while IHSPC_SJID+Z which use 3-D hyperspectral reflectance and elevation feature can bring a further improvement. The feature-level fusion method, DGF, has a better performance than HSPC_ori+Z, which directly assigns original spectra to LiDAR; however, it is still worse than intrinsic properties. The 3-D classification maps of different methods are shown in Fig. 12, where the proposed IHSPC_SJID+Z gives the best visual perception.

By comparing methods IHSPC_SJID, IHSPC_SJID+Z, IHSPC_SJID+I, and IHSPC_SJID+N, we can analyze the influence of different intrinsic properties, reflectance, height,

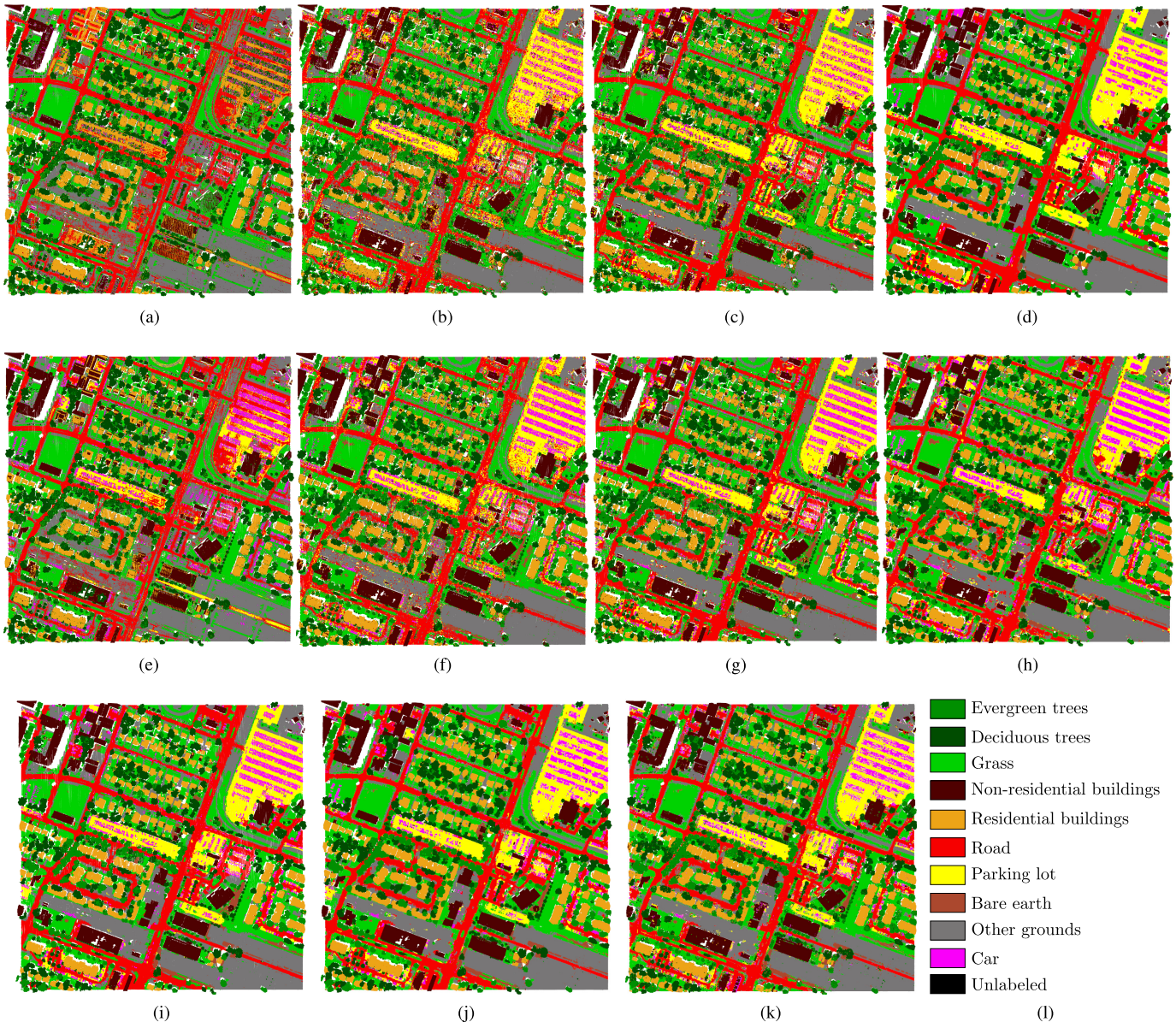

Fig. 12. 3-D classification maps of different methods. (a) MSPC. (b) HSPC_ori. (c) HSPC_2d_ref. (d) IHSPC_SJID. (e) MSPC+Z. (f) HSPC_ori+Z. (g) HSPC_2d_ref+Z. (h) DGF (i) IHSPC_SJID+Z. (k) IHSPC_SJID+I. (l) IHSPC_SJID+N.

intensity, and normal to classification performance. In general, we find that reflectance and height are the most useful properties for the classification task. Combining reflectance with intensity or normal, however, would worsen the classification performance than using only reflectance. The reason is that intensity is poorly influenced by noise, while the normal of points in the non-smooth surface like trees varies greatly, which results in poor performance in these categories. Normal is more useful for other applications such as segmentation, where it can be used to find smooth surfaces.

V. Conclusion

In this article, we have presented an SJID model for HSIs and LiDAR, which can effectively and efficiently recover intrinsic scene properties like 3-D hyperspectral reflectance and light direction. Our model can successfully handle an HSI which has high spectral dimensionality and a LiDAR point cloud that has unorganized data structures. Compared with other fusion methods of HSIs and LiDAR, the advantages of the proposed method is that it is based on the physical imaging principle other than human-designed features, and as a result, can generate IHSPC which is illumination-invariant and robust to shadows and misalignments. Moreover, the supervoxel-based optimization framework greatly reduces the computation complexity and makes it possible to handle millions of points within a reasonable time. We have also demonstrated the application of the proposed method in 3-D land cover classification. What is more, as a solution for combining HSIs and LiDAR into spectral 3-D data, our model can be easily applied to process data acquired by airborne or unmanned-aerial-vehicle-based (UAV-based) HSIs and LiDAR systems.

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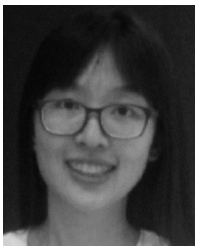

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